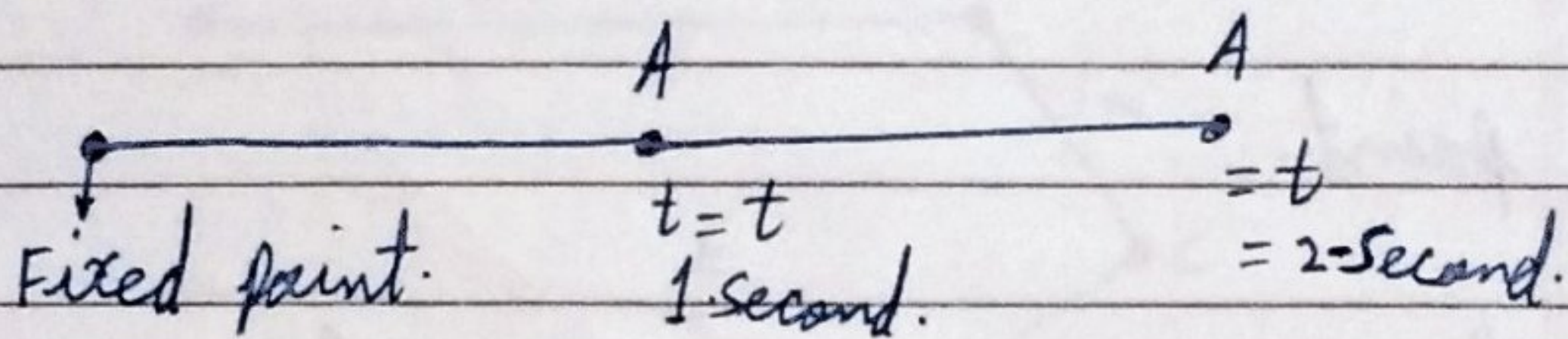
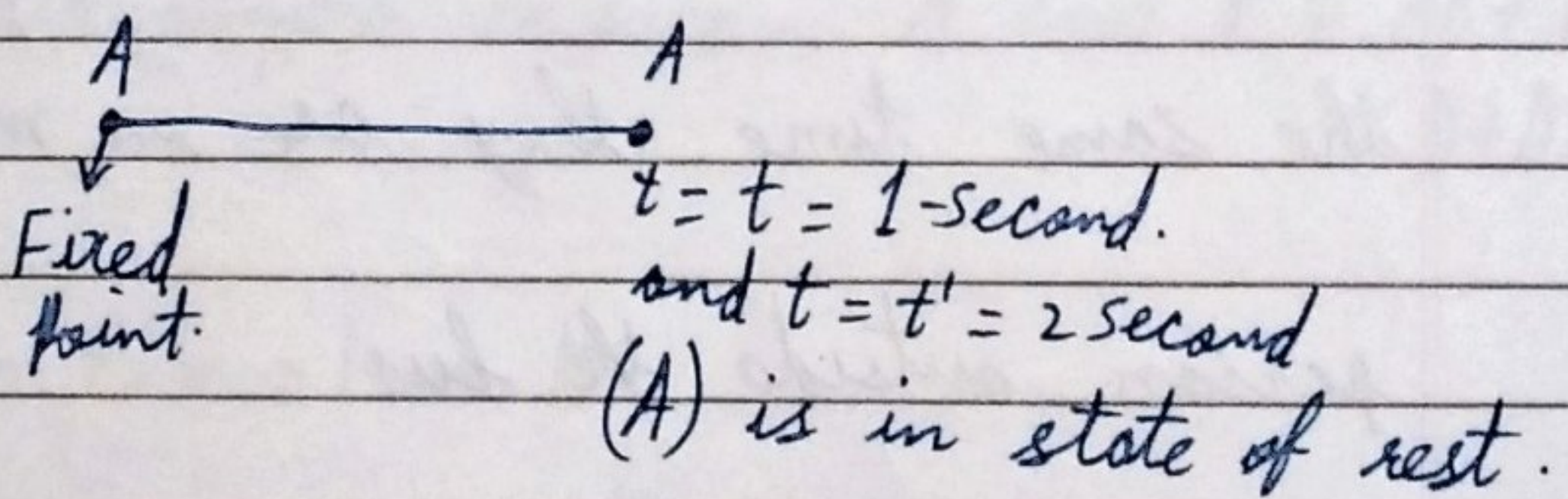


Motion

→ Motion:- A body is said to be in state of motion if it changes its position with respect to time and some fixed point.
(Reference point).



• Rest:-



- A body is said to be in state ^{of rest} ~~rest~~ if it does not change its position with respect

to time and some fixed point.

Note:- Motion and rest are relative to each other.

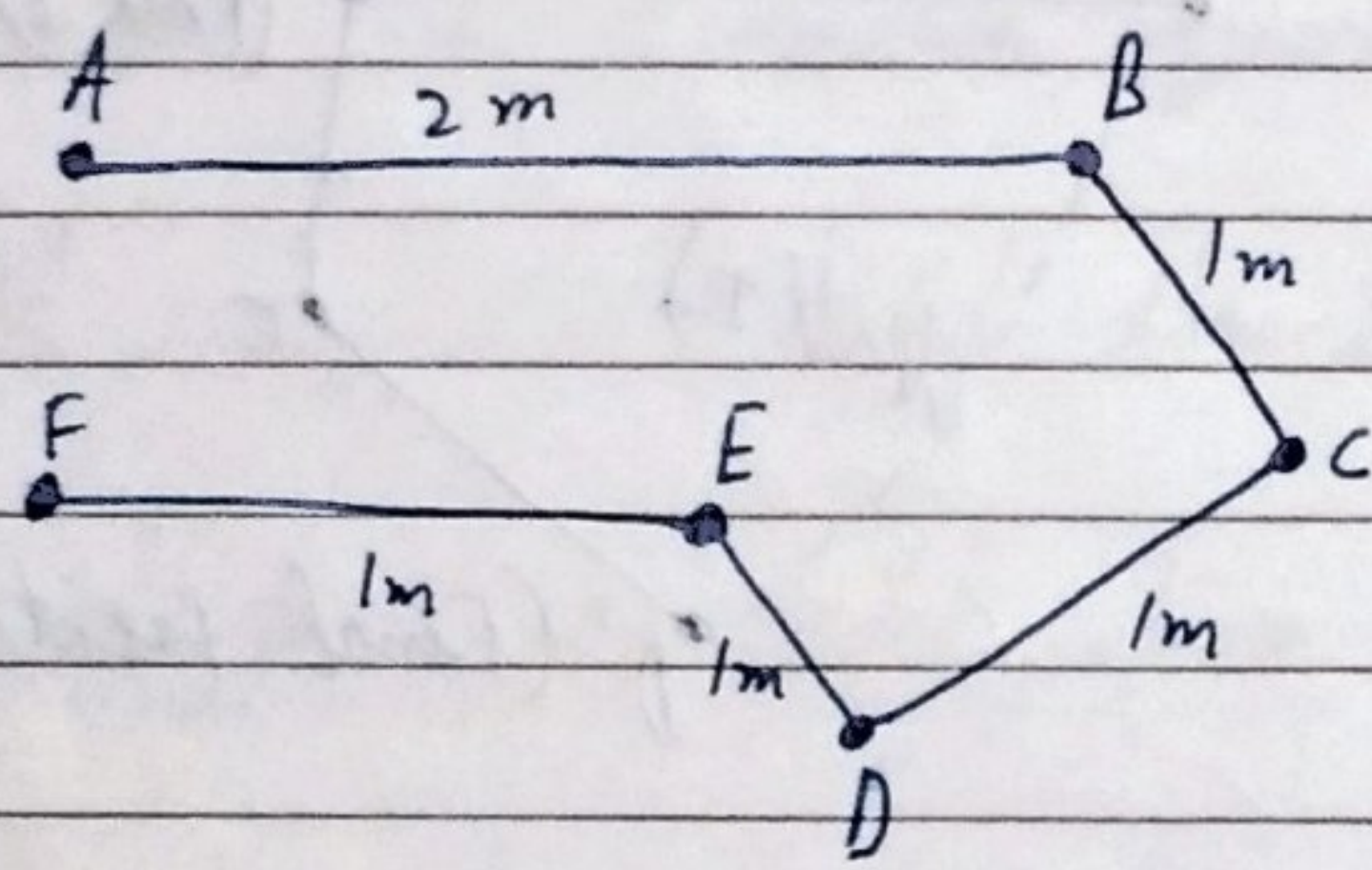
i.e At some time a body may be in state of rest or in motion depends upon reference point.

e.g Passengers in a moving bus are in state of rest with respect to each other but at the same time they are in motion for a person outside the bus.

→ Physical quantities:-

→ Distance:- It is also denoted by "s" or "d".

- Total path length covered by a body in a given time interval is called distance.



- Total path length between A and F = $AB + BC + CD + DE + EF$

$$= 2m + 1m + 1m + 1m + 1m$$

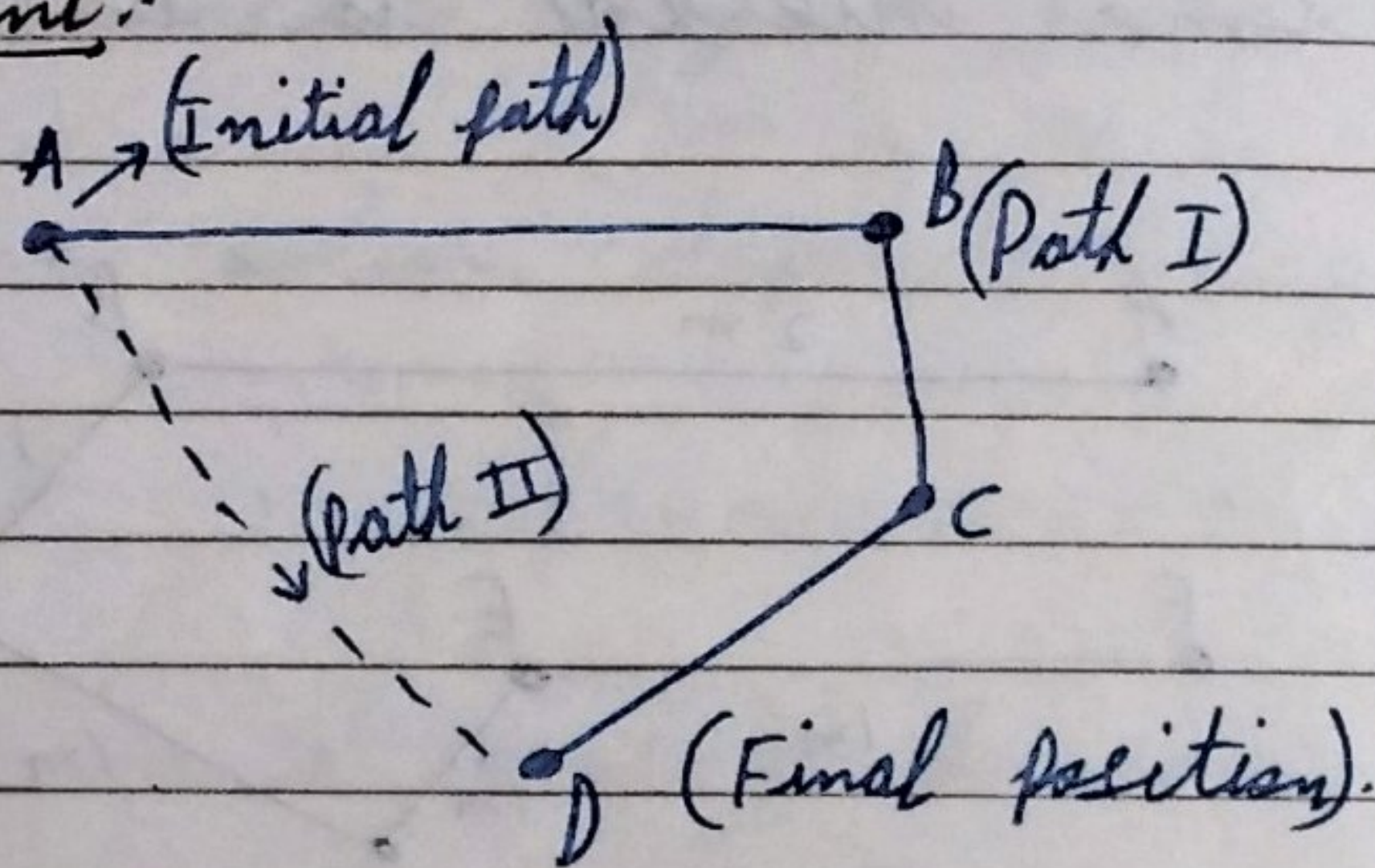
$$= 6m.$$

∴ Distance travelled = 6m.

→ Distance is a scalar quantity.

→ Scalar quantity: The physical quantity which have only magnitude and no sense of direction. e.g. distance. (Numerical value)

→ Displacement:



Distance covered by the body between A and D

$$= AB + BC + CD$$

∴ Displacement = AD

☆. Distance > Displacement.

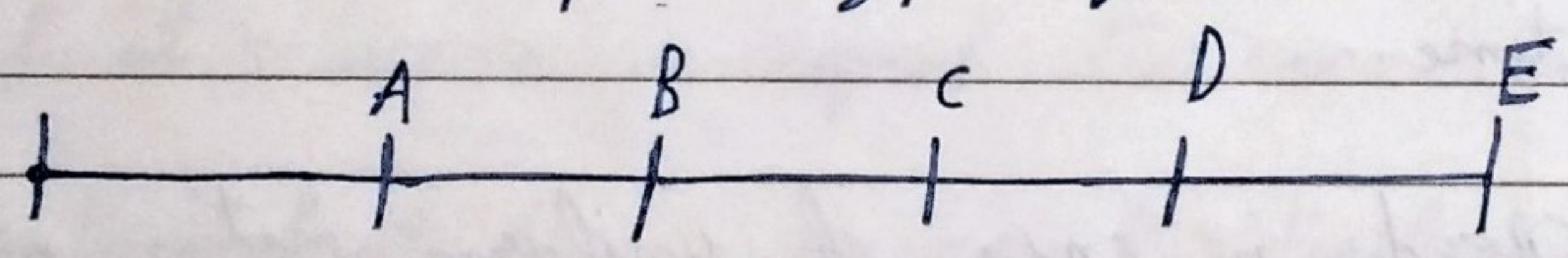
☆. Distance = Displacement. (Straight line).

→ On a circular path:-

- Displacement = 0
- Distance $\neq 0$
- Distance is always positive.
- Displacement is a vector quantity.
i.e both magnitude and direction.

→ Motion along a straight line:- (Rectilinear motion)

- It is the simplest type of motion.



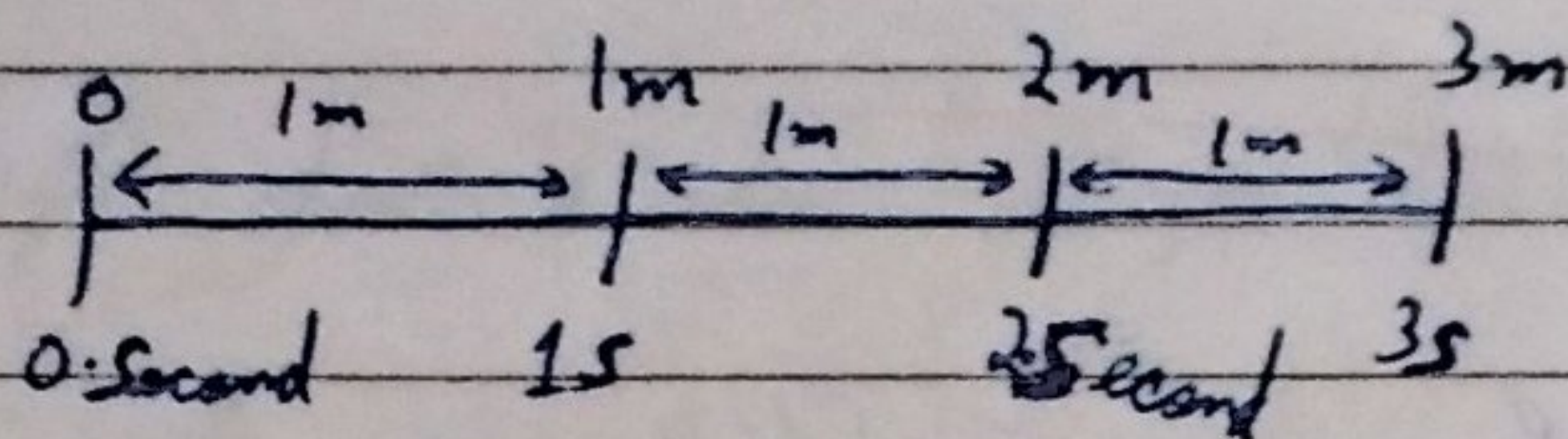
Displacement = distance.

e.g. A car moving on a straight road.

- An aeroplane moving on a runway.

- Types of the Motion:-

1) Uniform motion:-



- A body is said to be in uniform motion

~~eg*~~ if it covers equal displacements in equal intervals of time

~~eg*~~ Speed $\div$ Distance covered by the body in unit time.

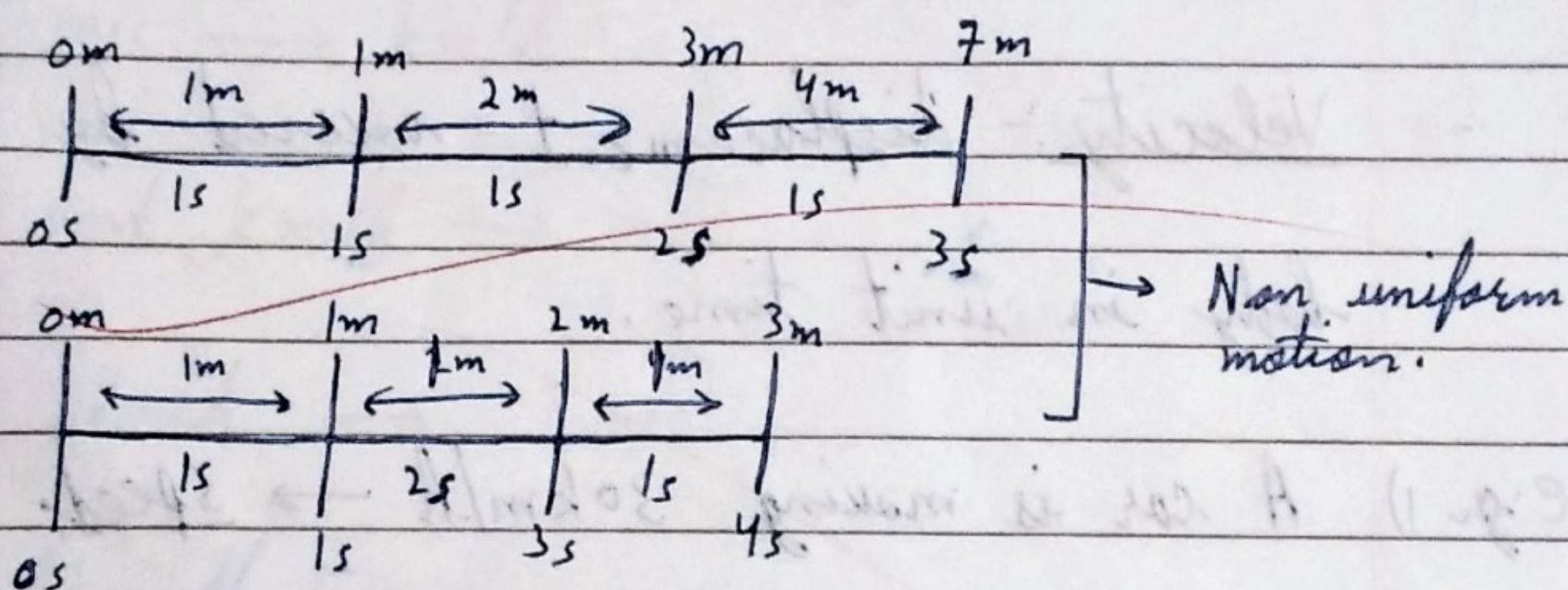
Speed in case of uniform motion ~~motion~~ is constant.

$$\text{Speed} = 1\text{m/s} \text{ or } \frac{3\text{m}}{3\text{Second}} = 1\text{m/s}.$$

*- Speed is a scalar quantity.

e.g. A bus travelling on a road with constant speed.

→ Non-uniform motion:-



- A body is said to be in non uniform motion if it covers unequal distances in equal interval of time or equal displacement in unequal intervals of time.

e.g. A bus moving with a variable speed.

- Average speed:- $\frac{\text{Total distance}}{\text{Total time}}$
- Speed with direction:- Velocity.
- Velocity:- displacement covered by a body in unit time.

e.g 1) A car is moving 30 km/h $\rightarrow$ Speed.

② A car is moving 30 km/h due east $\rightarrow$ Velocity.

Q:- An object travels 16 m in 4 seconds and then another 16 m in 2 seconds. What is the average of the object?

Sol We know,

$$\text{average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{16+16}{4+2}$$

$$= \frac{32}{6} = 5.33 \text{ m/s.}$$

- Change in units:

- $\text{m/s} \rightarrow \text{km/h}$

and $\text{km/h} \rightarrow \text{m/s}$

① $1 \text{ km/h} \rightarrow ? \text{ m/s}$

$$1 \text{ km} = 1000 \text{ m}$$

$$1 \text{ h} = 60 \times 60 = 3600 \text{ second.}$$

$$1 \text{ km/h} = \frac{1000}{3600} = \frac{5}{18} \text{ m/s.}$$

② $1 \text{ m/s} = ? \text{ km/h}$

$$1 \text{ m} = \frac{1}{1000} \text{ km}$$

$$1 \text{ s} = \frac{1}{3600} \text{ h}$$

$$1 \text{ m/s} = \frac{1}{\frac{100}{360}} = \frac{3600}{1000} \text{ km/h} \approx 3.6 \text{ km/h}$$

- Average velocity:

$$\star \rightarrow \text{Average velocity} = \frac{u + v}{2}$$

u = initial velocity.

v = final velocity.

Q $\rightarrow$

Sol

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

$$= \frac{90 + 90}{60} = \frac{180}{60} = 3 \text{ m/s}$$

-

$$\text{Average velocity} = \frac{\text{total displacement}}{\text{total time}}$$

$$= \frac{0}{60} = 0 \text{ m/s}$$

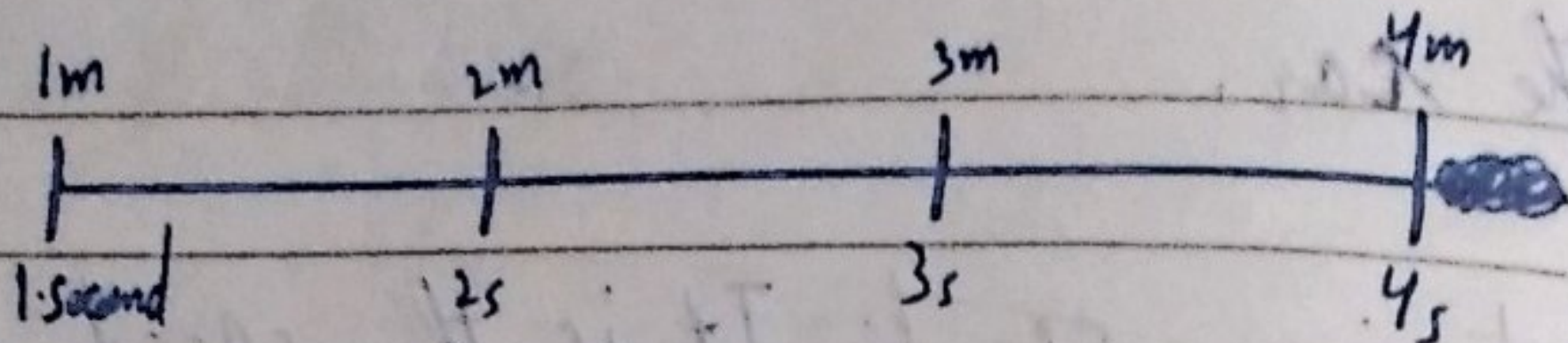
- Odometer: Distance travelled.
- Speedometer: It measures the instantaneous speed of the car.
- Instantaneous speed: It is the speed of the car at a instant of time.
- The odometer of a car reads 2000 km at the start of a trip and 2400 km at the end of the trip. If the trip took eight hour. Calculate the average speed of the car in km/h and m/s

Sol. Average speed = $\frac{\text{Total distance}}{\text{Total time}} = \frac{400}{8} \text{ km} = 50 \text{ km/h.}$

- Average speed in m/s = $50 \times \frac{5}{18} = \frac{125}{9} = 13.8 \text{ m/s.}$

- Change in velocity: (1)

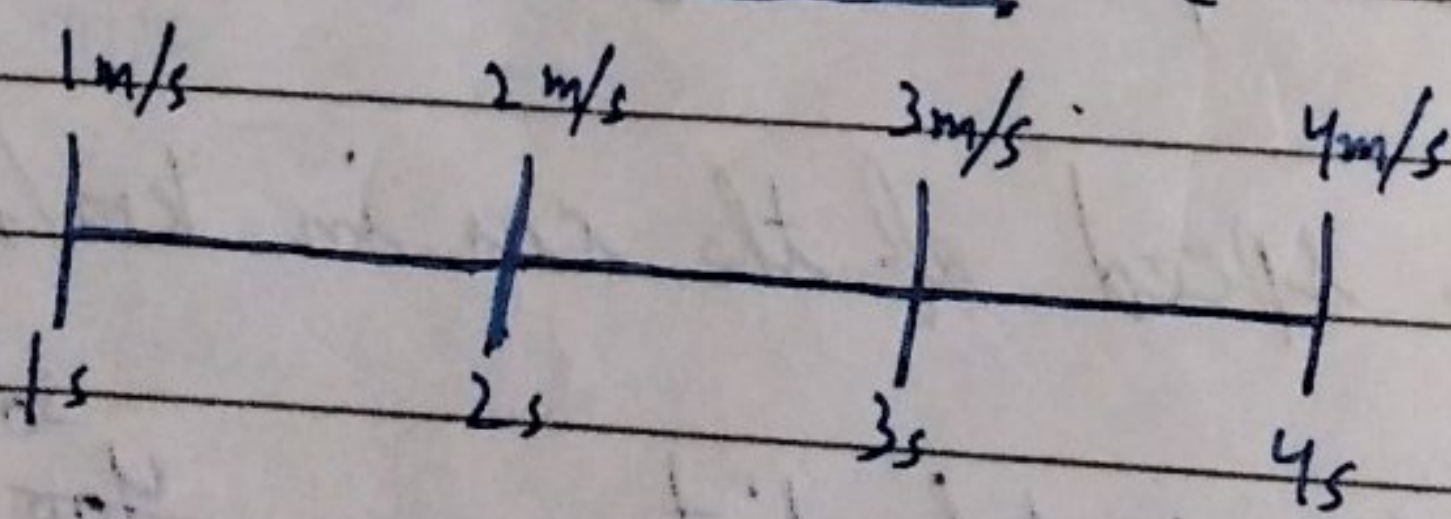
1) Uniform motion: In case of uniform motion



- The change in velocity is zero. [velocity is constant].

$\therefore$ - ~~Acceleration is zero in uniform motion.~~

2) - Non-uniform motion: (velocity is changing).



→ Velocity is changing at uniform rate

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time}} = \frac{V - U}{t}$$

$$\therefore a = \frac{V - U}{t}$$

- Acceleration :- $\frac{V-u}{t}$ $\left[\begin{array}{l} V = \text{final velocity} \\ u = \text{initial velocity} \end{array} \right]$

→ Acceleration :- Rate of change in velocity per unit time.

* - Velocity unit = m/s

* - Acceleration = $\frac{V}{t} = \frac{\text{m/s}}{\text{s}}$

$$a = \frac{\text{m}}{\text{s}^2}$$

Q. Starting from ^m a stationary position

Asim paddles his bicycle to attain a velocity of 6 m/s in 30 seconds. Then he applies brakes

such that the velocity of the bicycle comes

down to 4 m/s in the next 5 seconds.

Sol Speed = 40 km/h
in 10 minutes.

$$\Rightarrow \frac{40 \times 5}{10} = 11 \text{ m/s}$$

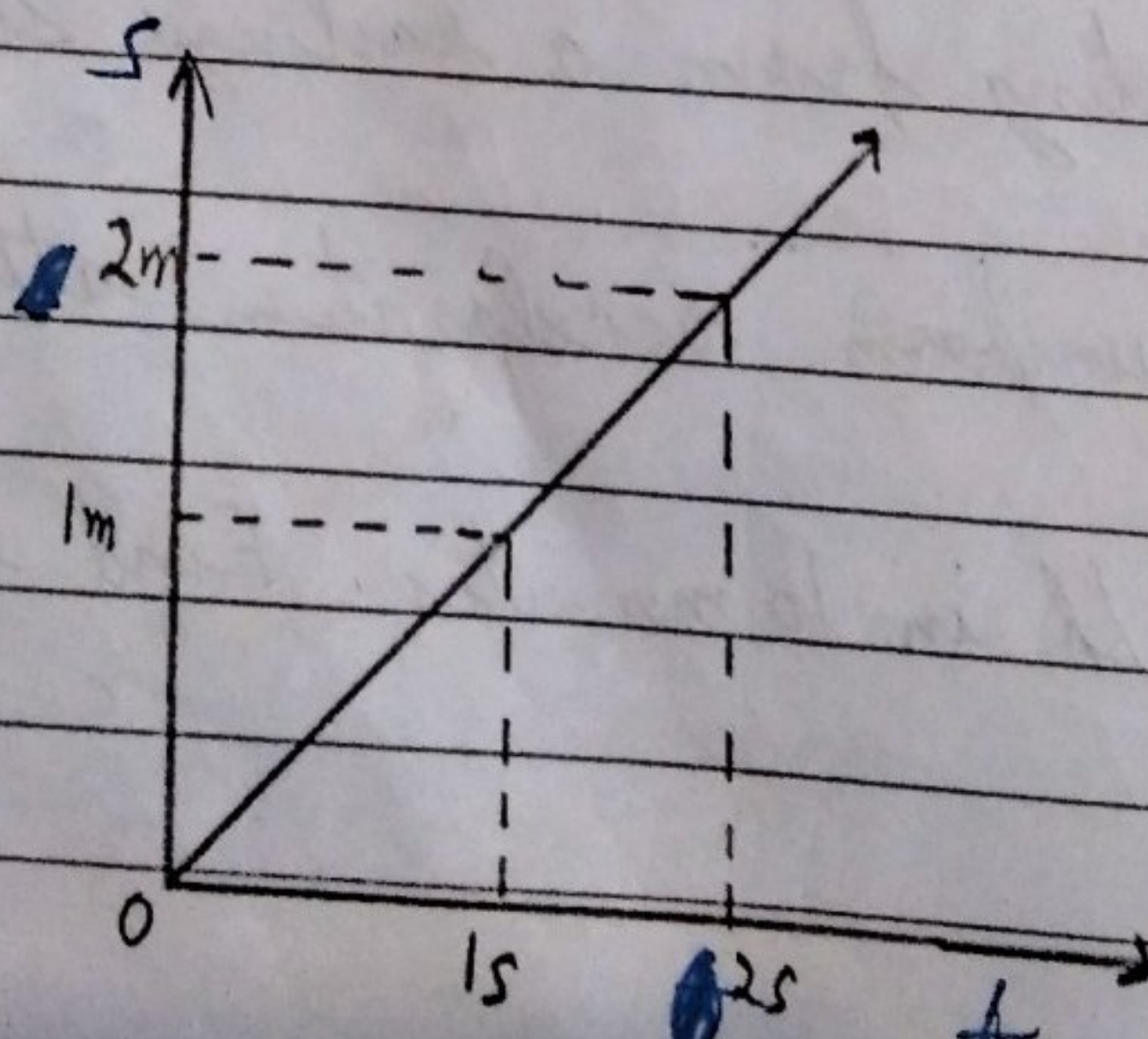
$$\Rightarrow 10 \times 60 = 600 \text{ s}$$

$$- a = \frac{v-u}{t}$$

$$a = \frac{11-0}{600} = \frac{11}{600} = 0.02 \text{ m/s}^2$$

→ Graphical representation of motion:

1 → Distance time graph of a body in uniform motion
(S.T. Graph).



$$V = \frac{S_2 - S_1}{t_2 - t_1}$$

graph is displacement.

→ Equations of motion:-

—————→ straight line motion
with uniform acceleration.

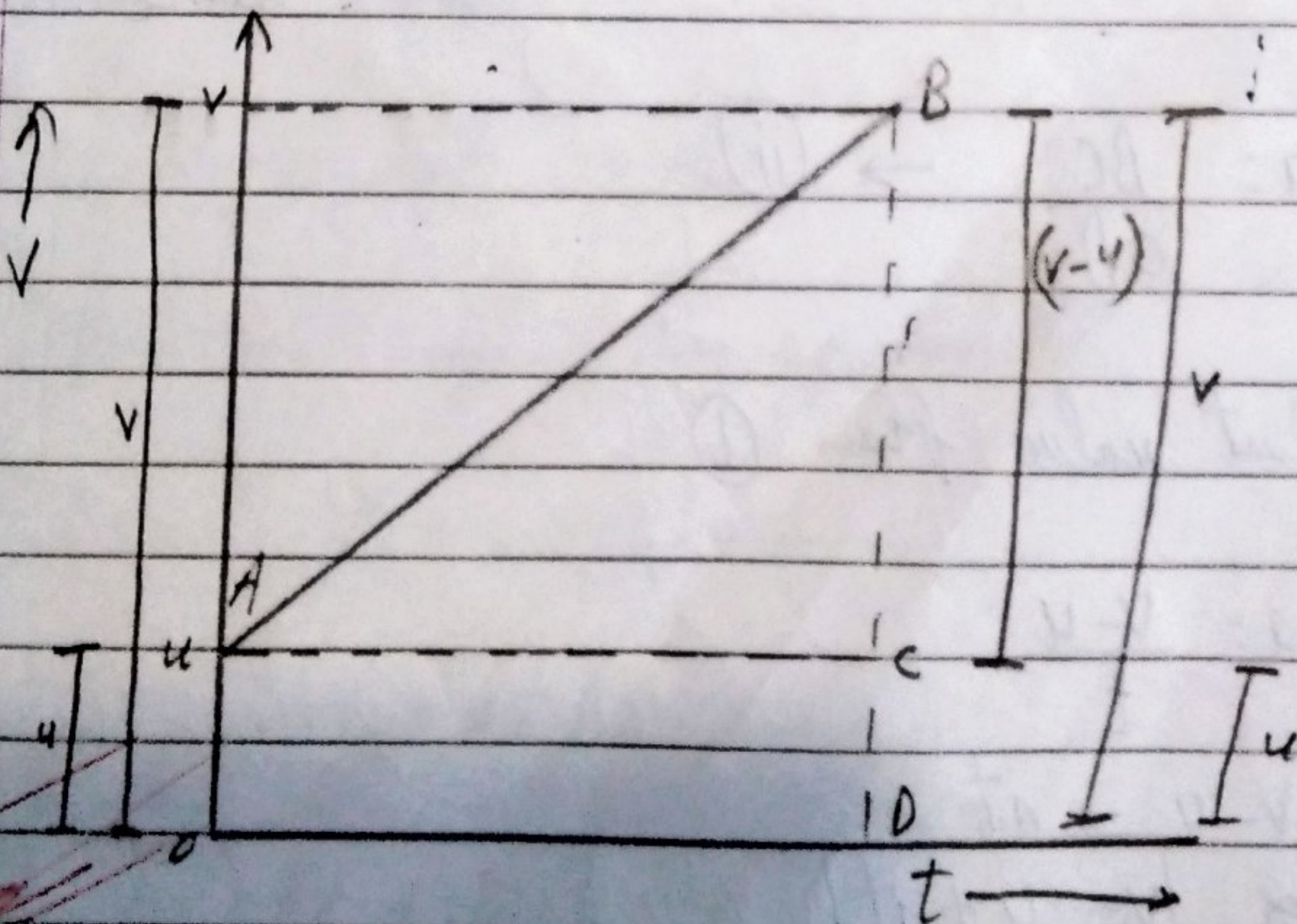
→ Three equation of motion.

1) $v = u + at$

2) $s = ut + \frac{1}{2} at^2$

3) $v^2 - u^2 = 2as$

— Equations of motion by graphical method:-



$$\Rightarrow OD = AC = t$$

$$OA = CD = u$$

$$BD = v$$

Now,

$$BC = BD - CD$$

$$\therefore BC = v - u \rightarrow \textcircled{1}$$

Also,

Acceleration: $a = \frac{\text{change in velocity}}{\text{time}}$

$$a = \frac{BC}{OD} \rightarrow \textcircled{ii}$$

Put value from $\textcircled{1}$

$$a = \frac{v - u}{t}$$

$$\therefore v - u = at$$

$$\text{or } \boxed{v = u + at}$$

- Second equation of motion:

As we know - Area under $v \cdot t$ graph = displacement.

$$\Rightarrow S = \text{Area of OABD (Trapezium)}$$

~~$$S = \frac{1}{2} (\text{sum of // sides}) \times \text{height.}$$~~

$$S = \frac{1}{2} (OA + BD) \times AC$$

$$S = \frac{1}{2} (u + v) \times t \rightarrow \textcircled{I}$$

- Also we know,

$$v = u + at \rightarrow \textcircled{II}$$

Put value of (v) in equation $\textcircled{I}$.

$$\Rightarrow S = \frac{1}{2} (u + u + at) \times t$$

$$S = \frac{1}{2} (2u + at) \times t$$

$$\therefore S = \left(\frac{1}{2} \times 2u + \frac{1}{2} at \right) t$$

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$$S = \left(u + \frac{1}{2} at \right) t$$

$$\boxed{S = ut + \frac{1}{2} at^2} \rightarrow \text{Displacement.}$$

→ Third equation of motion:

- S : Area under graph.

- S = Area of trapezium (OABD).

$$S = \frac{1}{2} (\text{Sum of // sides}) \times \text{height.}$$

$$S = \frac{1}{2} (OA + BD) \times AC$$

$$S = \frac{1}{2} (u + v) \times t \rightarrow (1)$$

- We know: $v = u + at$

$$v - u = at$$

$$\boxed{t = \frac{v - u}{a}}$$

→ Put the value of (t) in equation (1).

$$S = \frac{1}{2} (v + u) \left(\frac{v - u}{a} \right)$$

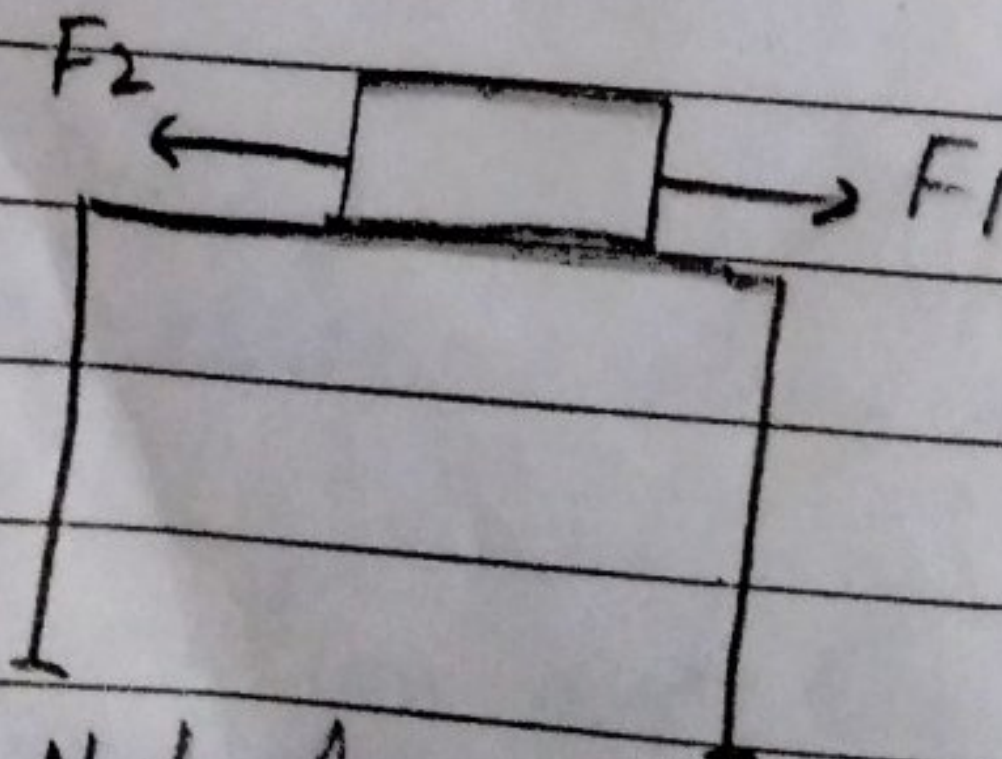
Force and Laws of Motion

→ Inertia: The property by virtue of which a body is unable to change the state of rest or motion.

→ Force: It is an external effort or Agent in the form of Push or Pull which changes the state of the body. (Either motion or rest).

→ Two types: (1) Balanced and unbalanced force.
(2) ~~balanced~~ ~~to~~ Unbalanced force.

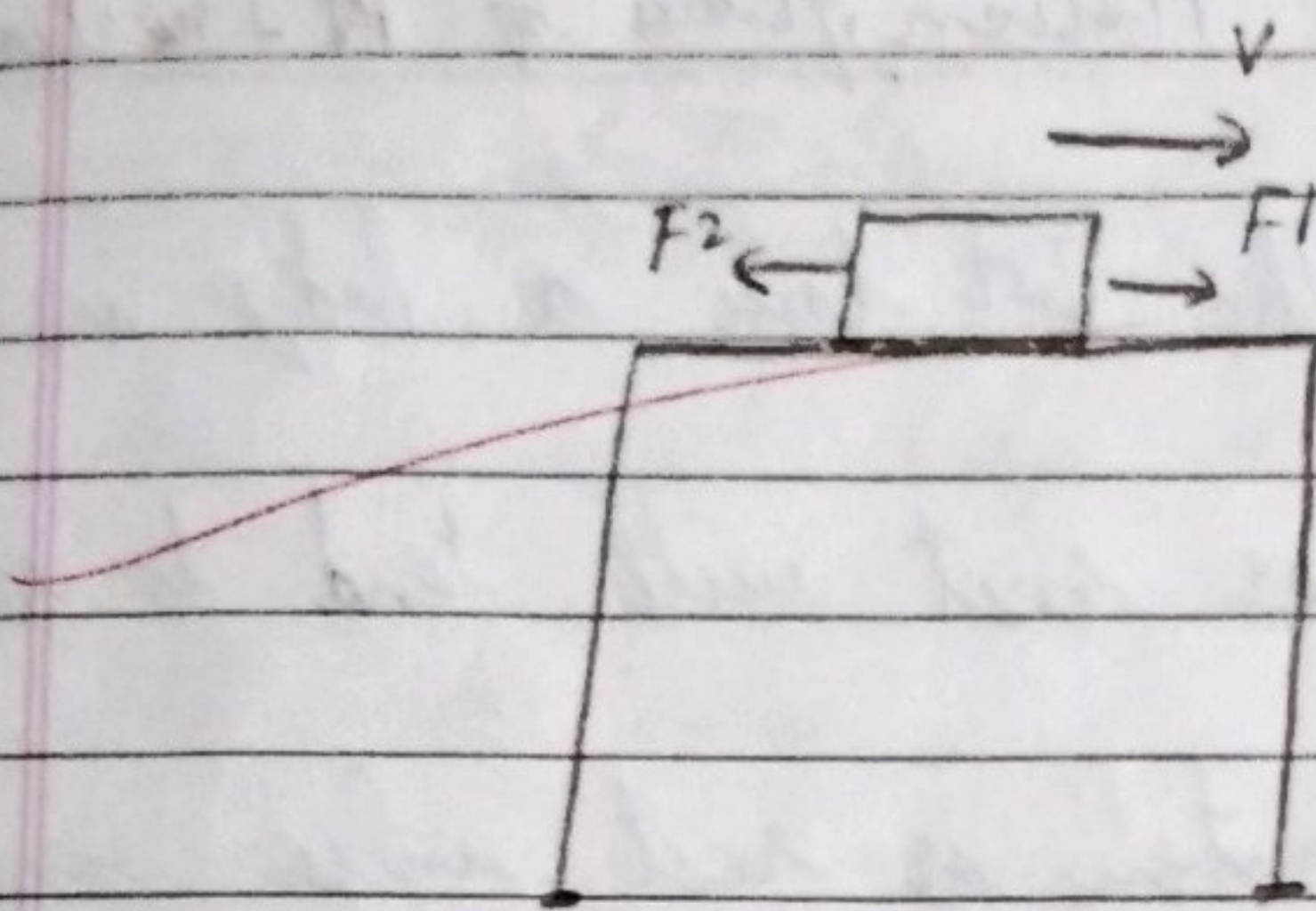
(1) Balanced force:



∴ $F_1 = F_2$ - Net force = 0.

→ If the resultant of all the forces acting on a body is zero, then this force is called as balanced force.

(2) Unbalanced Force:-



$$\therefore \boxed{F_1 > F_2} - \text{Net force} \Rightarrow F_1 - F_2$$

Body will move in the direction of F_1 .

e.g. If $F_1 = 10\text{ N}$
 $F_2 = 7\text{ N}$

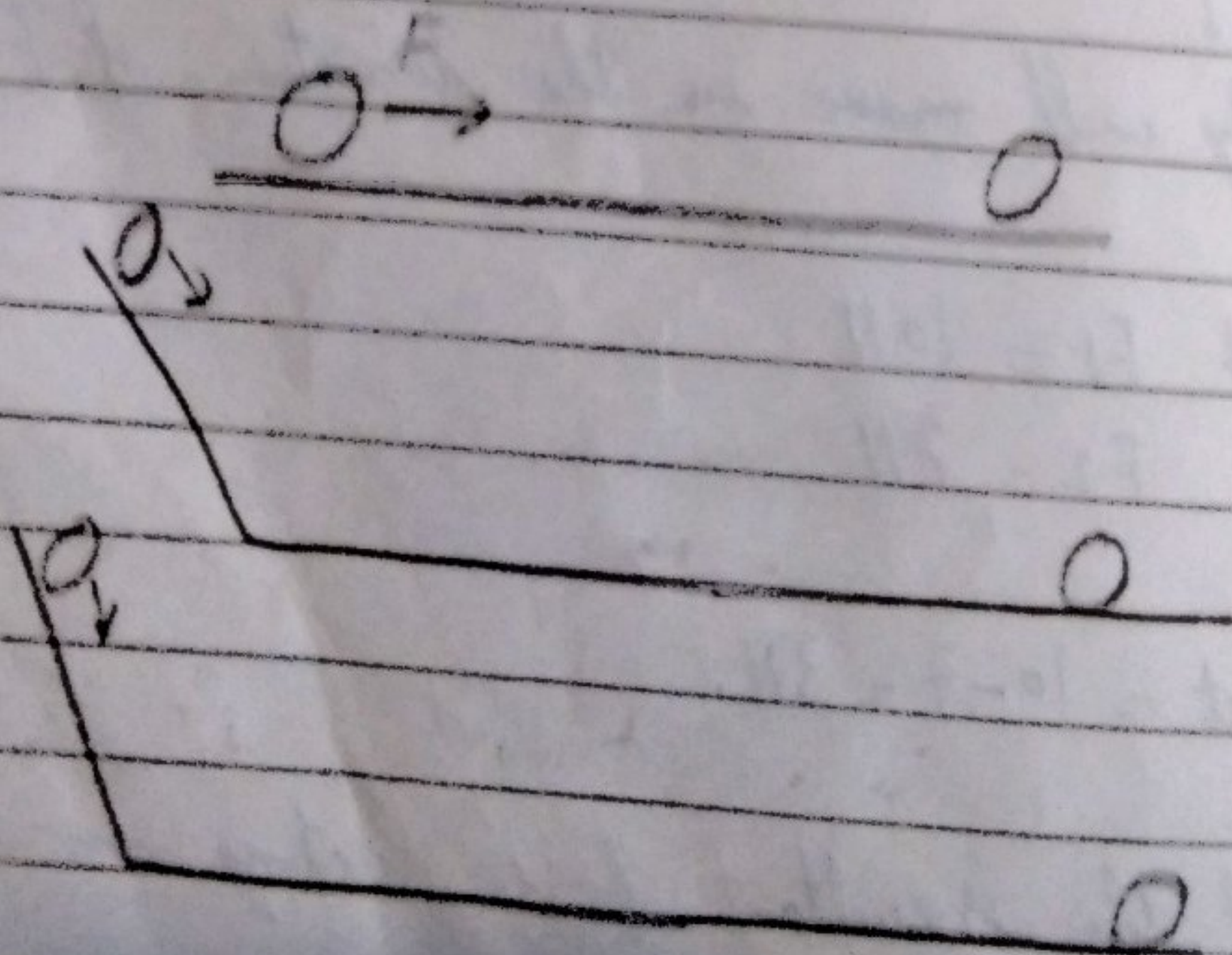
$$F_{\text{Net}} = 10 - 7 = 3\text{ N.}$$

→ If the resultant force acting on a body is not

equal to zero, then the forces acting on a body are called as ~~unb~~ unbalanced forces.

① $\Rightarrow$ First law of Motion: / Law of Inertia

- According to the 1st law a body in state of a motion or rest will tend to remain in state of motion or rest unless some external agent did not act on it.



u

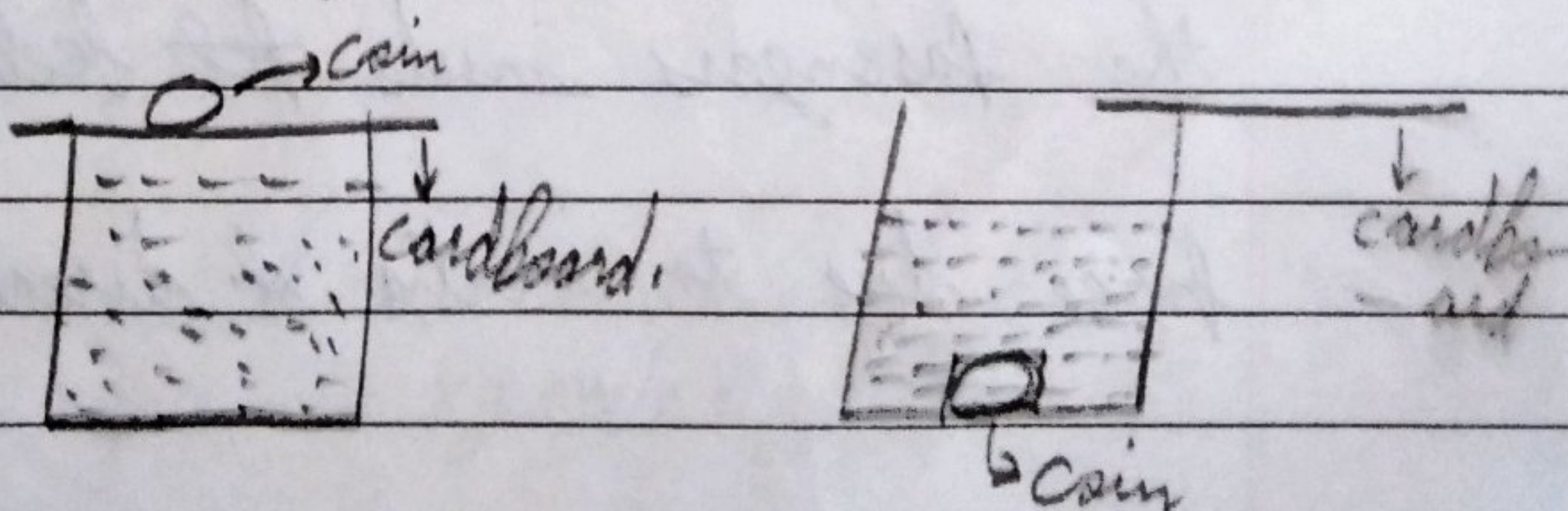
→ Inertia It is the property through which body resists any change in its state of motion or rest

→ Types of Inertia ~

1) Inertia of Motion It is a type of Inertia due to which body tends to remain in the state of motion.

e.g. Passengers in a moving bus fell forward when brakes are applied suddenly, due to inertia of Motion.

2) Inertia of Rest:-



→ It is a type of inertia in which a body tends to remain in state of rest.

e.g. Upon striking a cardboard the coin placed on it fell in the beaker due to inertia of rest. While cardboard moves in forward direction.

(3) Inertia of direction: It is a type of inertia in which the body ~~opposes~~ in the change in the direction of motion.

e.g. While taking a curve ~~via~~ via moving bus, the passengers inside ~~for~~ feel an outward force due to inertia of direction.

→ Momentum:

$$P = m \times v$$

if $v = 0$ (rest)

$$P = 0$$

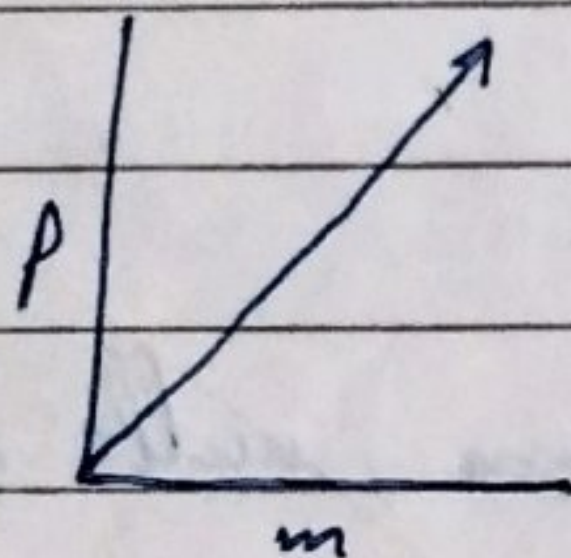
$$mv = 0$$

$$m \neq 0$$

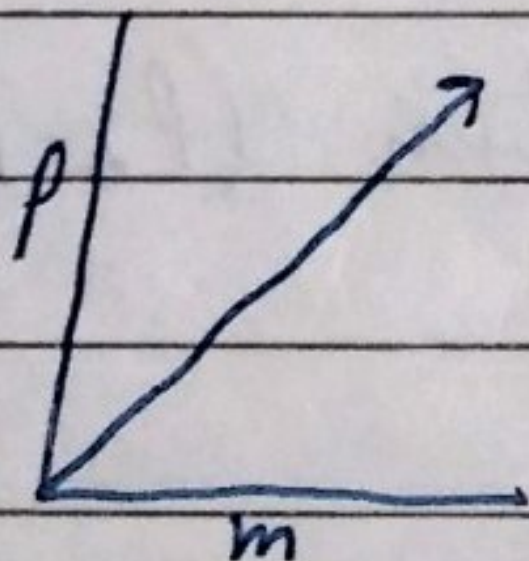
⇒ ∴ It is a product of mass and velocity.

~~$v = \text{constant}$~~

~~$P \propto m$~~



$m = \text{constant} \therefore P \propto v$



⇒ ~~$m = \text{const}$~~ Unit: $P = m \times v$
 $= \text{kg} \times \text{m/s} = \text{kgm/s}$

(2) → Newton's 2nd law of motion: The force applied to the body is directly proportional to the change in the momentum.

→ Let initial momentum is $p_1 = mu$
where (u) is initial velocity.

- Now a constant force F is applied for (t) time.

∴ The final momentum will be $p_2 = (mv)$

Where (v) is final velocity

∴ Change in ~~was~~ momentum $(p_2 - p_1)$

- According to 2nd law

$$\Rightarrow F(t) \propto (p_2 - p_1)$$
$$F(t) \propto mv - mu$$

$$F(t) \propto m(v-u)$$

$$\therefore F \propto \frac{m(v-u)}{t}$$

$$\Rightarrow F \propto ma$$

$$\text{or } F = Fma$$

$\hookrightarrow$ Constant.

$$\therefore \boxed{F = ma}$$

$\star \rightarrow$ Unit of Force:

$$m = \text{kg}$$

$$a = \text{m/s}^2$$

$$\boxed{F = \text{kg m/s}^2 = \text{Newton}}$$

$\Rightarrow$ Deduction of 1st law from 2nd Law:

According to ^{2nd} ~~1st~~ law :-

$$F = ma$$

$$\text{if } F = 0$$

$$\therefore ma = 0$$

$$m \neq 0 \Rightarrow$$

$$\boxed{a = 0}$$

$$\text{or } \frac{v-u}{t} = 0$$

$\therefore$ The body will move with uniform velocity which is first law

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⇒ Applications of 2nd law: *

1) A Cricket player lower down his hands while taking a catch, because this increases the time duration for change in momentum, hence the impact of the force will be less.

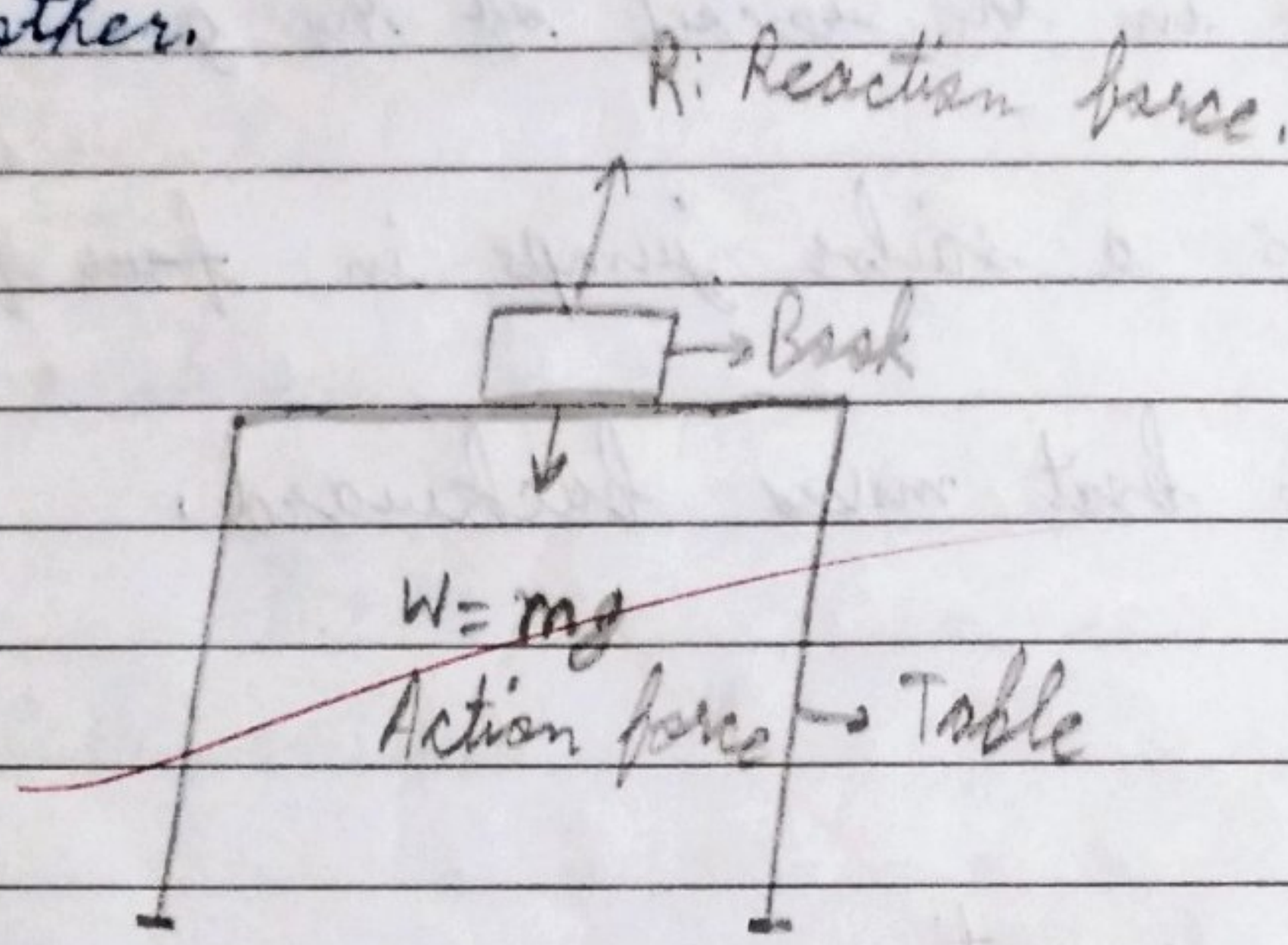
2) Vehicles are fitted with shock absorbers to increase the time duration for changing change in momentum.

(3) ⇒ Newton's 3rd law of motion:

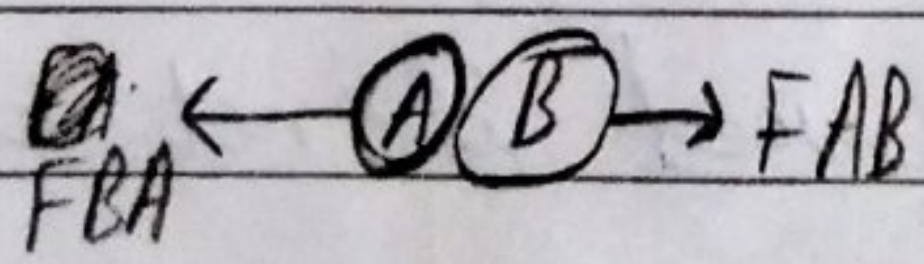
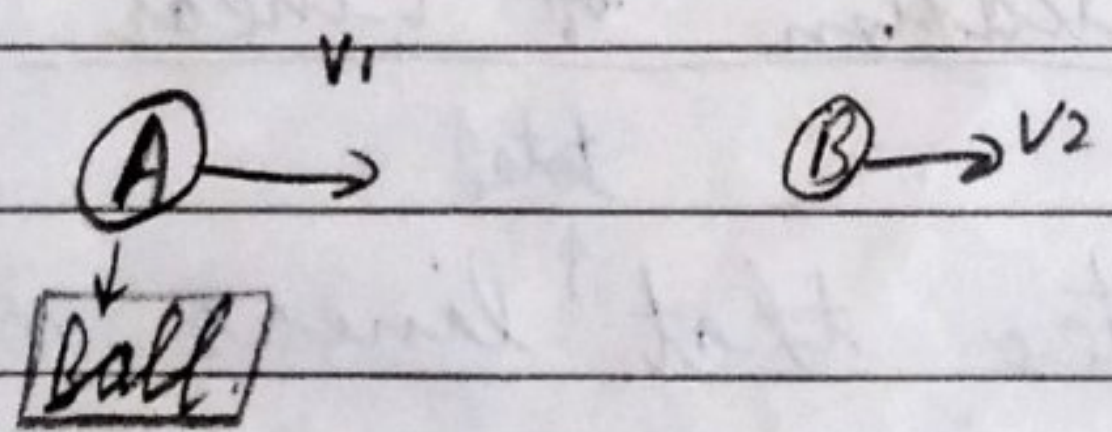
- To every action there is equal and opposite reaction. i.e.; Action and reaction

forces are equal in magnitude and opposite in direction.

- Action and reaction forces acts upon two different objects so they never cancel with each other.



$\therefore R = mg = W$



Acc. to Newton's 3rd law: $F_{AB} = -F_{BA}$

-ive sign indicates opposite direction.

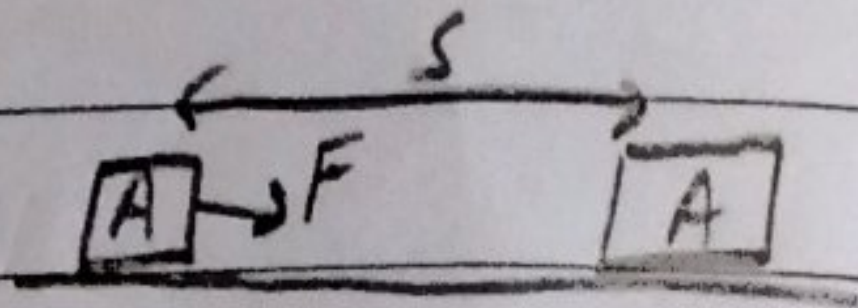
Work, Energy and Power

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⇒ Work: Work is said to be done when we apply force and it displaces the body in the direction of force.



☆ Work done = $F \times S$

→ Work done is equal to the product of force and displacement.

☆ → Unit of Work done: $W = FS$
 $= N \times m = \text{Joule}.$

⇒ S.I unit of Work done = Joule.

Q) A force of 10N. is applied on a body and it displaces the body ~~by~~ by 2 metre in the

amount.

direction of force. Calculate the ¹ Work done.

Ans Force = 10 N.
Displacement = 2 metre

• Work done = Force $\times$ Displacement.

$$W = 10 \times 2$$

$$W = 20 \text{ Joule.}$$

$\Rightarrow$ Work done in different cases:-

1) Positive Work done: If the direction of the displacement and force is same then work done is said to be positive work done.

2) Negative Work done: If the direction of displacement and force is opposite to each other then work done is said to be negative.

Q1). A force of 7N acts on a body the displacement is say (8m) in the direction of force. What is the work done in this case?

Ans Given,

$$\text{Force} = 7\text{ N}$$

$$\text{Displacement} = 8\text{ m}$$

$$\therefore \text{Work done} = \text{Force} \times \text{displacement}$$

$$W = 7 \times 8$$

$$W = 56 \text{ joule.}$$

⇒ Zero Work done.

⇒ If $s = 0$

$$\therefore \text{Work done} = F \times 0 = 0$$

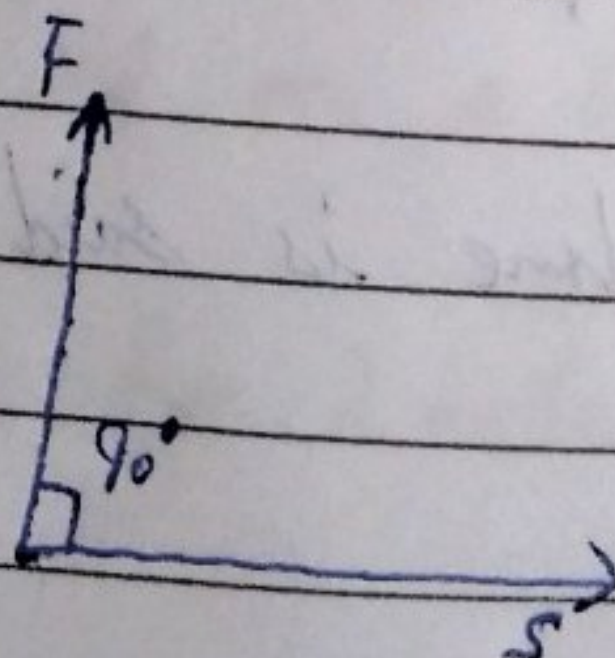
e.g. A boy pushes a concrete wall.

⇒ Work done by a coil

$$W = F s \cos \theta$$

$$\theta = 90^\circ \quad \cos 90^\circ = 0$$

$$\boxed{W = 0}$$



Q A porter lifts a luggage of 15 kg from the ground and puts it on his head 1.5 m above the ground. Calculate the work done by him on luggage.

Ans $W = F \times S$
 $m = 15 \text{ kg} \cdot S = 1.5$
 $F = m \times g = 15 \times 10 \text{ m/s}^2 = 150 \text{ N}$
 $\therefore W = F \times S = 150 \times 1.5 = 225 \text{ joule.}$

* $\Rightarrow$ Energy: Capacity to do Work is called energy.

$\Rightarrow$ Types of Energy:

1) Kinetic Energy

2) Potential Energy.

1. $\Rightarrow$ Kinetic Energy:- The energy due to the motion of the body is called K.E energy.

eg. Energy possessed by the running water.

$$\Rightarrow k.E_i = \frac{1}{2} m u^2 = \frac{1}{2} \times 1500 \times \left(\frac{150}{18}\right)^2$$

$\Rightarrow$ Types of Energy-

1) Potential Energy.

2) Kinetic Energy.

1) Kinetic Energy - The energy by virtue of the motion in a body is called as kinetic energy.

e.g. A moving car has K.E.
A moving Fan
Flowing, etc.

$\Rightarrow$ Expression for the K.E-

Let us consider a force (F) acts on a body.

$$\therefore F = ma \quad \text{--- (1)}$$

∴ From the third equation of motion

$$v^2 - u^2 = 2as$$

$$\therefore \boxed{a = \frac{v^2 - u^2}{2s}}$$

$$\therefore \text{Work done } W = Fs$$

$$= mas \quad [F = ma]$$

$$W = m \left(\frac{v^2 - u^2}{2s} \right) s = \frac{1}{2} m (v^2 - u^2)$$



If $u = 0$

$$\boxed{W = \frac{1}{2} m v^2} = \boxed{K.E}$$

→ K.E is never negative it is always Positive

because it depends upon magnitude of Velocity

i.e., Speed

e.g. An Object of mass 15 kg is moving with

a uniform Velocity of 4 m/s. What is the KE

possessed by the object?

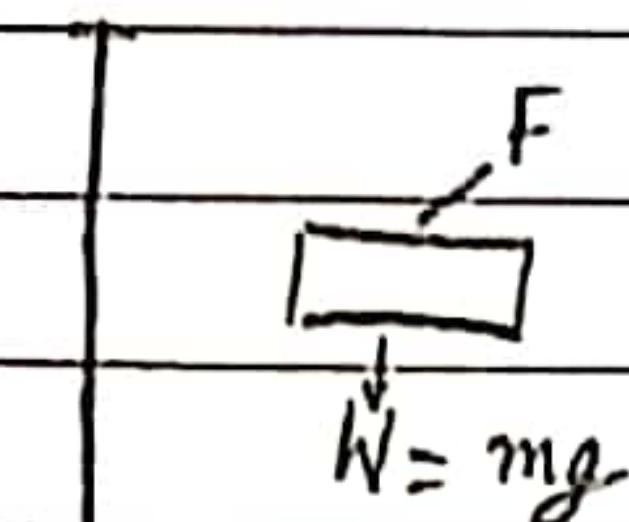
Sol

$$m = 15 \text{ kg}$$
$$V = 4 \text{ m/s}$$

$$\begin{aligned} \text{K.E} &= \frac{1}{2} mV^2 = \frac{1}{2} \times 15 \times (4)^2 \\ &= \frac{15}{2} \times 16 = 120 \text{ Joule.} \end{aligned}$$

2) Potential Energy:- The energy possessed by an object by virtue of its position, i.e; height from the ground.

⇒ Expression for Gravitational P.E.



Gravity

∴ Workdone against to gravity

$$W = F \times s = F \cdot h$$

$$F = mg$$

$$W = mgh = U$$

⇒ U depends on (h)

- Different height = ~~Different~~ Different P.E
- P.E is path independent.

⇒ Power: Power is defined as the rate of doing work or. Workdone per unit time

$$P = \frac{W}{t}$$

- Unit = $\frac{J}{s} = \text{Watt}$.
- S.I unit of power = Watt.
- Define one Watt. $1W =$ if $W = 1J$
and $t = 1s$

∴ one Watt is defined as when one Joule of Work

is done in one second.

Commercial Unit of Power

- ~~Horse~~ Horse power. (hp)

$$\Rightarrow 1 \text{ h.p.} = 746 \text{ Watts.}$$

$$\boxed{\frac{1 \text{ h.p.}}{2} = \frac{746}{2} = 373 \text{ Watts}}$$

Bigger Units of Power:

$$\bullet 1 \text{ Kilowatt} = 1 \text{ kW} = 10^3 \text{ W}$$

$$\bullet 1 \text{ Megawatt} = 1 \text{ MW} = 10^6 \text{ W} = 1000000 \text{ W}$$

$$\bullet \text{Energy: } W = P \times T = \frac{\text{J.s}}{\text{s}} = \text{J.}$$

Commercial Unit of energy:

$$\bullet \boxed{\text{Kilowatt hour} = \text{kWh}} = \text{Unit}$$

- $1 \text{ kWh} = 1 \text{ unit.}$

- Energy consumed in 1h at the rate of 1 kilowatt.

- ~~1 kWh~~ $1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s}$
 $= \boxed{3600000 \text{ W.s} = 3.6 \times 10^6 \text{ J}}$

Q. An Electric bulb of 60 Watt is used for 6 hrs per day. Calculate the units of energy consumed in one day by the bulb.

Sol) $P = 60 \text{ W} = \frac{60}{1000} = 0.06 \text{ kW.}$

$t = 6 \text{ hrs.}$

- $E = \text{kWh} = 0.06 \times 6$
 $= 0.36 \text{ kWh.}$
 $= 0.36 \text{ units.}$

Q. An object of mass 12kg. is at a certain

height above the ground. If the potential energy of the object 480 Joules. Find the height. Take

$$g = 10 \text{ m/s}^2$$

Sol. Given,

$$m = 12 \text{ kg}$$

$$g = 10$$

$$\therefore P.E = mgh = 480$$

$$mgh = 480$$

$$h = \frac{480}{mg} = \frac{480}{12 \times 10} = 4 \text{ m.}$$

Q- An Electric heater is rated 1500 Watt. How much energy does it use in 10hr.

Sol: $P = 1500 \text{ W}$

$$t = 10 \text{ hrs.}$$

$$\frac{1500}{1000} = 1.5 \text{ kWh.}$$

$$E = \text{kWh} = 1.5 \times 10 = 15 \text{ kWh} = 15 \text{ units.}$$

- In terms of Energy

15 x 1 kWh:

$$15 \times 3.6 \times 10^6 \text{ J} = \boxed{54 \times 10^6 \text{ J}}$$

Q) A certain household has consumed 250 units of energy during a month. How much energy is this in Joules.

sol)

250 x 1 kWh

250 x 3.6 x 10⁶ J

$$= \boxed{900 \times 10^6}$$